TECHNICAL & COMMERCIAL QUOTATION

OPTION 2 – ONE CENTRALISED DEHUMIDIFIER SERVING THE THREE COLD ROOMS

Quotation Ref.	MES/QTN/HVAC/ESM/2026-200	**Date**	27 August 2026
Client	**M/s EBN SINA MEDICAL**	**Attention**	Mr. Yasser Al Attar
Project	**Desiccant Dehumidification of Three (3) Cold Rooms**	**Validity**	30 days from date of offer

Subject: Supply & Installation of Desiccant Dehumidifier for Three (3) Cold Rooms – Option 2: Single Centralized Unit

Dear Mr. Yasser,

With reference to your kind enquiry and our subsequent discussion regarding humidity control inside your cold storage rooms, we are pleased to submit herewith our technical and commercial proposal for the supply and installation of a centralised desiccant (adsorption) dehumidification system serving the three (3) cold rooms.

This offer covers Option 2, based on one single centralised dehumidifier serving all three cold rooms through a ducted distribution network. A separate offer (Ref. MES/QTN/DEH/ESM/2026-206) is submitted in parallel covering Option 1, based on a dedicated unit per cold room.

1. PROPOSED SOLUTION

We propose to supply and install one (1) no. AIRDRY model AD 1250 adsorption dehumidifier, manufactured by Tecnofrigo Tuscany Srl (TFT), Italy, installed centrally in the adjacent plant / ambient area and serving all three cold rooms through a common ducted distribution network.

- Process air is drawn from the three cold rooms through a common return duct, dried across the silica-gel rotor, and the dry air is distributed back to the three rooms through an insulated supply duct network fitted with balancing dampers.
- The regeneration (wet) air stream is ducted and discharged to atmosphere, outside the treated space.
- A single wall-mounted mechanical humidistat controls the operation of the unit; the dry air distribution between the three rooms is set by balancing dampers at commissioning stage.
- The unit is of a higher capacity than the sum of three individual units, providing a reserve of capacity for peak loading and heavy door traffic.

Points to be noted: the three rooms are served from one common system, so all three rooms operate at a common humidity control regime, and the whole system is out of service during maintenance of the unit. The option also requires a distribution ductwork network between the plant location and the three rooms.

2. SCOPE OF WORK

No.	Description
1	Supply & Installation of 1 no. AIRDRY AD 1250 adsorption dehumidifier (TFT – Italy), complete with ADKMH2 mechanical wall-mounted humidistat.
2	Testing, commissioning, setting of the required RH set-point and verification of performance in each room.

3. TECHNICAL PARTICULARS – AIRDRY AD 1250

EQUIPMENT IDENTIFICATION	
Equipment	Adsorption (desiccant) dehumidifier, silica-gel rotor type
Model	AIRDRY AD 1250
Brand / Manufacturer	TFT – Tecnofrigo Italy or TROTEC GERMANY
Country of Origin	Italy or Germany
Quantity	1 No. (centralised, serving 3 cold rooms)
PERFORMANCE	
Dehumidification Capacity	5.54 kg/h at 20 °C / 60% RH
Process Air Flow	1,250 m³/h
Process Operating Temperature Range	–20 °C to +40 °C
ELECTRICAL	
Power Supply	400 V / 3 Phase + N / 50 Hz
Installed Regeneration Heater Power	6.60 kW
Process Fan Nominal Power	440 W
Reactivation Fan Nominal Power	175 W
Rotor Drive Motor	3.0 W
Maximum Power Absorbed	**7.22 kW**
Maximum Current Absorbed	**13.6 A**
DIMENSIONS, WEIGHT & CONNECTIONS	
Dimensions (A × B × C)	581 × 745 × 740 mm
Empty Weight	62 kg
Process Air Inlet	Ø 250 mm
Dry Air Outlet	Ø 250 mm
Regeneration Air Inlet	Ø 160 mm
Wet (Regeneration) Air Outlet	Ø 160 mm
CONTROL ACCESSORY	
Humidistat	ADKMH2 – mechanical wall-mounted humidistat, control range 30 – 90% RH, 1 step, IP54 (1 No. included)

4. COMMERCIAL OFFER

No.	Description	Qty	Unit	Rate (QAR)	Amount (QAR)
1	Supply, delivery and installation of AIRDRY AD 1250 centralised adsorption dehumidifier (TFT – Italy), complete with ADKMH2 mechanical humidistat, support frame, wet air discharge duct with weather	1	No.	91,000.00	91,000.00

No.	Description	Qty	Unit	Rate (QAR)	Amount (QAR)
	louvre, control wiring, final connection, testing, commissioning and handover.				
2	Supply and installation of the insulated dry air supply and process air return duct distribution network serving the three cold rooms.	1	Lot	Included	Included
	TOTAL LUMP SUM PRICE – OPTION 2				**91,000.00**

7. COMMERCIAL TERMS & CONDITIONS

Payment Terms	70% advance payment against Purchase Order | 30% against completion of installation and commissioning
Delivery Period	3 – 4 weeks from date of receipt of the advance payment
Installation Period	Within 5 – 7 working days from date of delivery to site, subject to site readiness
Warranty	12 months manufacturer's warranty on the equipment against manufacturing defects from the date of delivery, plus 12 months warranty on our installation workmanship.
Validity of Offer	30 (thirty) days from the date of this quotation
Currency	Qatari Riyals (QAR). Prices are for supply and installation at site, Doha – State of Qatar

We trust the above meets with your kind approval and remain at your disposal for any technical clarification or for a joint site visit to finalise the installation arrangement.

We thank you for the opportunity to quote and look forward to your valued order.

Yours faithfully,

Metri Engineering Services W.L.L.

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AIRDRY

Adsorption Dehumidifiers

AD 270÷2200

TFT

DRY AIR SOLUTIONS

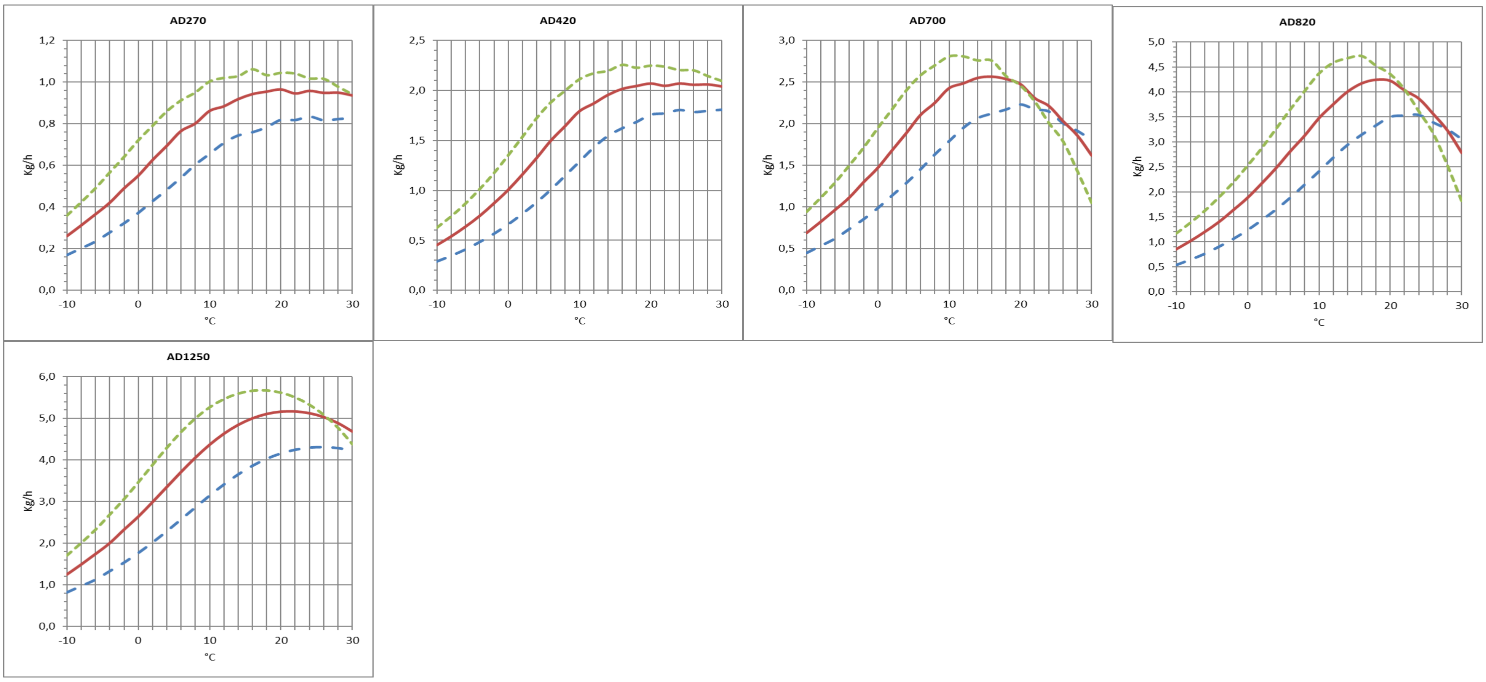
TECHNICAL DATA								
MODEL	**AD**	**270**	**420**	**700**	**820**	**1250**	**1700**	**2200**
Performances								
Dehumidification Capacity *	Kg/h	0,99	2,17	2,58	4,32	5,54		
Fans								
Process air flow	m^3/h	270	420	700	820	1250		
Static pressure	Pa	340	360	300	100	230		
Fan nominal power	W	138	157	170	170	440		
Reactivation air flow	m^3/h	50	90	135	210	270		
Static pressure	Pa	135	100	130	210	130		
Fan nominal power	W	85	175	175	175	175		
Drive Motor								
Nominal power	W	3,0	3,0	3,0	3,0	3,0		
Regeneration								
Regeneration type		Electric	Electric	Electric	Electric	Electric		
Installed power	KW	1,75	2,6	3,5	6,6	6,6		
Temp. rise in the heating coil	°C	75	85	90	105	85		
Electrical characteristics								
Power supply	Volt/Ph/Hz	230/1/50	230/1/50	230/1/50	400/3+N/50	400/3+N/50		
Maximum power absorbed	KW	1,97	2,93	3,84	6,95	7,22		
Maximum current absorbed	A	6,6	12,7	12,9	12,6	13,6		
Noise level								
Sound pressure **	dB (A)	47	49	51	64	69		
Sound power **	dB (A)	75	77	79	92	97		

* *Conditions at 20°C 60% RH*
** *Sound pressure level calculated in free field, 1 meters from unit, direction factor Q = 2, according to ISO 9614*

DEHUMIDIFICATION CAPACITY

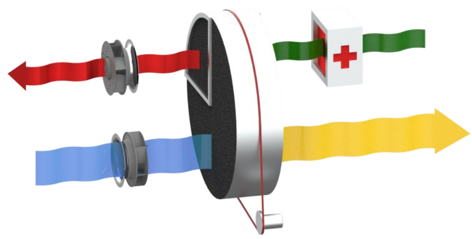
Approximate capacity in Kg/h with different relative humidity values of inlet process air (RH%).

40% RH 60% RH 80% RH



PRINCIPLE OF OPERATION

The dehumidifier operates using two distinct airflows: the main one consists of the air to be dehumidified, while a second, smaller airflow is used for rotor regeneration. Two fans inside the unit generate these airflows, which pass through the rotor in opposite directions. The process air, meaning the air to be dehumidified, flows through the desiccant rotor impregnated with silica gel, a highly hygroscopic material capable of absorbing the water vapor present in the air. As it passes through, the air releases moisture to the rotor, which retains it. The dehumidified air is then directed into the production area or the process requiring treatment.
The dehumidification process takes place within a temperature range of -20°C to +40°C, which is specific for the correct functioning of the rotor but not as an operational limit of the unit. During the cycle, the rotor rotates slowly, driven by a transmission system with a gear motor and belt. The regeneration air, used to remove the moisture absorbed by the rotor, is heated by an internal coil up to approximately +100°C. This air passes through the rotor in the opposite direction to the process air, causing the release of the moisture accumulated by the rotor and restoring its absorption capacity. The regeneration air, now warm and humid, is expelled outside the treated environment.



STRUCTURE

The structure of the dehumidifier is made of galvanized steel and stainless steel to ensure resistance and long-term durability. The simple panel design, without thermal insulation, provides a robust and compact construction. The top panel is easily removable, allowing quick and convenient access to electrical components and internal mechanical parts for maintenance operations. For installation, the dehumidifier is compatible with standard spiral ducts, simplifying the connection to the ventilation system.

FANS

The fans are directly coupled to a single-phase AC or EC motor, with IP55 protection rating, class F insulation, and class B efficiency. Maintenance access is quick and easy thanks to the inspectable top panel. When the dehumidifier starts, both the process fan and the regeneration fan immediately begin operating.

ROTOR

The dehumidifier is equipped with an adsorption rotor made of high-performance desiccant material. Its honeycomb structure, composed of heat-resistant corrugated sheets, integrates silica gel as the desiccant material, ensuring a high moisture absorption capacity in a compact volume. This design optimizes airflow, creating a large number of axial fluid streams that maximize the efficiency of the dehumidification process. The rotor is designed to withstand saturated air without damage and can tolerate sudden stoppages of the process or regeneration fan without compromising its operation. Additionally, it is non-combustible and non-flammable, ensuring maximum operational safety.

TRANSMISSION SYSTEM

The rotor's movement is managed by a belt-driven transmission system, ensuring smooth and efficient operation. The belt applies traction to the outer edge of the rotor and is guided by a pulley connected to the gear motor. To ensure maximum reliability, a tensioning device keeps the belt in the correct position, preventing slippage and ensuring consistent operation over time. The rotation direction and proper functioning of the transmission can be easily checked by opening the top panel. The rotor is supported by ball bearings, which increase its lifespan and reduce friction, while its steel shaft provides strength and mechanical resistance even under intensive use conditions.

REGENERATION AIR HEATING COIL

The dehumidifier is equipped with a self-regulating PTC electric regeneration heater, designed to ensure efficiency and operational safety. Thanks to PTC (Positive Temperature Coefficient) technology, the heater can self-regulate its surface temperature, keeping it constant and preventing the risk of overheating. This ensures precise control of the regeneration process, improving the system's reliability and lifespan.

FILTERS

The dehumidifier is equipped with a G3 filter positioned at the inlet of both the process and regeneration airflows, designed to capture dust particles and air impurities. This filter ensures a clean airflow, helping to maintain the system’s operational efficiency and protecting internal components from potential damage caused by contamination.

ELECTRICAL PANEL

The electrical panel of the dehumidifier is designed and manufactured in compliance with European regulations 73/23 and 89/336, ensuring high standards of safety and efficiency. Access to the panel is simple and secure by removing the unit's top panel.
Each unit comes standard with a main switch, ammeter, hour counter, and a connector for connecting an external humidistat. Additionally, an energy meter can be installed as an option to monitor electrical consumption.
The electrical panel is also equipped with a switch for manual or automatic control of the dehumidification process. It allows configuring the continuous operation of the process fan, even when the desired humidity level has been reached, provided the external humidistat is connected and in automatic mode.

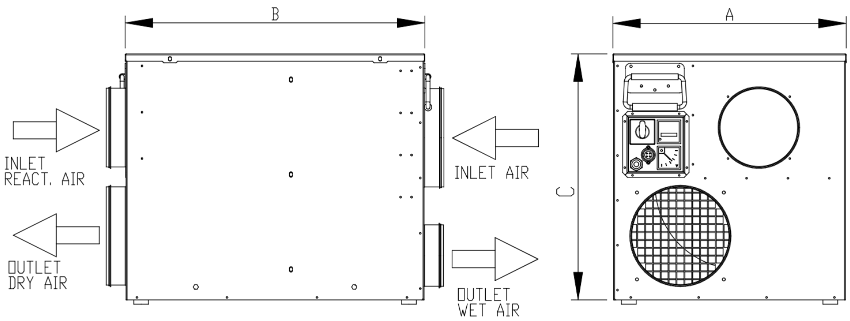
VERSION

AD… Standard
AD…/HR Version with heat recuperator for regeneration air (50%-80% recovery).

Model AD	Code	270	420	700	820	1250	1700	2200
Painted steel frame		○	○	○	○	○	○	○
Satin-finished304 stainless steel frame		●	●	●	●	●	●	●
Cover box for outdoor installation	ADKOPB	○	○	○	○	○	○	○
Regeneration air heat recovery	/HR	○	○	○	○	○	○	○
Turbo box for increase available head	ADKTBP	○	○	○	○	○	-	-
Power switch		●	●	●	●	●	●	●
G3 process and regeneration filters		●	●	●	●	●	●	●
Filters F5, F7, F9		-	-	-	-	-	-	-
Pre- and Post-treatment of Air		-	-	-	-	-	-	-
PLC electronic control and touch-screen terminal		-	-	-	-	-	-	-
Different supply voltage		○	○	○	○	○	○	○
Dirty process air filter warning	ALFP	-	-	-	-	-	-	-
Dirty regeneration air filter warning	ALFR	-	-	-	-	-	-	-
Electronic wall-mounted humidistat 1 step	ADKHW230+	○	○	○	○	○	○	○
probe from duct or wall relative humidity 0÷100%	ADKH1D/W	○	○	○	○	○	○	○
wall probe relative humidity 10÷95%	ADKH2W	○	○	○	○	○	○	○
Electronic wall-mounted 1-step humidistat with RH% probe	ADKHTW1	○	○	○	○	○	○	○
Mechanical humidistat from duct and wall 15÷95%	ADKMH1	○	○	○	○	○	○	○
Mechanical wall-mounted humidistat 30÷90%	ADKMH2	○	○	○	○	○	○	○

● standard, ○ optional, – unavailable.

Dimensions



Model	AD	270	420	700	820	1250	1700	2200
A	mm	407	475	475	581	581		
B	mm	450	609	609	745	745		
C	mm	428	509	509	740	740		
Empty weight	Kg	23	35	37	57	62		
Connections								
Process air inlet	mm	Ø 125	Ø 200	Ø 200	Ø 250	Ø 250		
Dry air outlet	mm	Ø 125	Ø 200	Ø 200	Ø 250	Ø 250		
Regeneration air inlet	mm	Ø 125	Ø 160	Ø 160	Ø 160	Ø 160		
Wet air outlet	mm	Ø 80	Ø 125	Ø 125	Ø 160	Ø 160		